

Sentiment Analysis of Twitter US Airline

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Project Overview

This project aims to identify sentiments in the US airline industry using tweets posted by travellers in Twitter in February 2015 in order to identify and classify travellers opinions by applying sentence-level sentiment analysis, where each tweet was analyzed individually and classified into positive, negative, or neutral sentiment categories based on tweet polarity. This will help in understanding travelers satisfaction, identifying common service issues, and improving overall customer experience.

Dataset Description

The dataset was used in this project is Twitter US Airline Sentiment dataset obtained from Kaggle. It includes travelers opinions about US airlines posted in Twitter platform in February 2015. This dataset usually used in sentiment analysis tasks in order to study travelers opinions expressed in text form. It contains 14,640 records where each record represents a single tweet posted by a traveler and labelled according to their sentiment as positive, negative, or neutral. Also, the dataset contains 15 features:

Variables	Data Type	Description
tweet_id	Numeric	Unique identifier assigned to each tweet
airline_sentiment	Object	Sentiment labelling for each tweet: positive, negative, or neutral
airline_sentiment_confidence	Numeric	Confidence score of the sentiment classification
negativereason	Object	Reason associated with negative tweets
negativereason_confidence	Numeric	Confidence score of the negative reason classification

airline	Object	Name of the airline
airline_sentiment_gold	Object	Human-verified sentiment label
name	Object	Twitter username of the user who posted the tweet
negativereason_gold	Object	Human-verified negative reason label
retweet_count	Numeric	Number of times the tweet was retweeted
text	Object	Actual tweet text written by the user
tweet_coord	Object	Geographic coordinates of the tweet
tweet_created	Object	Date and time when the tweet was posted
tweet_location	Object	Geographic location reported by the user
user_timezone	Object	Time zone associated with the user

Table 1. Variables description

Methodology

The sentiment analysis project was conducted using Python within a Jupyter Notebook environment. Several Python libraries were used to support sentiment analysis such as pandas, numpy, re, TextBlob, matplotlib.pyplot, and seaborn. The Twitter US Airline Sentiment dataset was loaded using `read_csv()` function. After successfully loading the dataset, the `'head()'` function was used to display the head of dataset to provide a brief preview of the dataset structure and understand its variables. Multiple exploratory steps were done to examine dataset and understand its structure and quality before applying data preprocessing and sentiment analysis techniques. First, shape attribute was used to extract the number of records and variables within dataset while `info()` function was used to display important information such as number of column and its names, the number of non-null values, and datatype for each variable. Also, `isnull().sum()` function was used to detect incomplete data per column. Then,

`duplicated().sum()` function was used to examine repeated tweets within the dataset.

`describe()` function was used to extract descriptive statistics for specific categorical variables including airline sentiment, airline name, and negative reason. Other variables were not included because they mainly contain identifiers, metadata that provide limited value for sentiment analysis exploration. Finally, `countplot()` function was used to explore the distribution of tweet sentiments as well as negative reasons associated with tweets. These visualizations are important because they help in understanding the frequency of sentiment categories within the dataset and identifying the most common customer complaints reported by airline travelers. After the exploratory phase, dataset undergoes several preprocessing techniques to prepare tweet text for sentiment analysis. First, duplicate records were removed using `drop_duplicates()` function to avoid repeated records that may affect the analysis results. Then, a custom function was created called `clean_twt()` to preprocess tweet text by removing mentions, URLs, hashtag symbols, numbers, special characters, and extra white spaces from tweets. To improve text consistency during sentiment analysis, all tweet text was converted to lowercase using `lower()`. Then, cleaned tweets were stored in a new column called `cleaned_tweet` while keeping the original tweet text in the `text` column. Finally, the cleaned dataset was re-checked by obtaining dataset shape, duplicate records, and the head of the dataset after preprocessing. In sentiment analysis phase, custom function called `estimate_sentiment()` was created in order to classify tweet sentiments based on sentiment polarity scores generated with help of the TextBlob library. The `TextBlob().sentiment.polarity` function was used to estimate the polarity of each cleaned tweet and classify it into positive, negative, or neutral sentiment categories. The Classification of sentiments based on the following:

- Polarity values greater than 0 were classified as positive

- Polarity values less than 0 were classified as negative
- polarity values equal to 0 were classified as neutral

After that, sentiment analysis was performed on the `cleaned_tweet` column using the `apply()` function, where the result of analysis for each tweet was stored in a new column called `estimated_sentiment`. After that, new dataframe was created and called `sentiment_analysis_df` to store only the cleaned tweet text and its associated estimated sentiment. Then, both columns were renamed to `Tweet` and `Associated_Sentiment` to make the final output easier to understand. Finally, the `head()` function was used to display the head of sentiment analysis dataframe.

Results

Dataset exploration showed that Twitter US Airline Sentiment dataset contained 14,640 tweet records with 15 features, where most of features were object variables. Several variables contained missing values, especially `airline_sentiment_gold`, `negativereason_gold`, and `tweet_coord` which had a large number of missing values. The `negativereason` and `negativereason_confidence` variables contained missing values since it mainly applicable to negative tweets, while positive and neutral tweets naturally should not be associated with negative reasons. Furthermore, 36 duplicate records were identified in the dataset before preprocessing. Descriptive statistics for some categorical variables showed that the dataset contained three unique sentiment categories where the negative sentiment was the most common category with frequency of 9178 which indicates large portion of traveller expressed dissatisfaction toward airline services. Also, it showed that there is six different US airlines where United airline appeared most frequently in the dataset with 3,822 tweets. Regarding

negative reasons, ten unique negative reasons were reported where “Customer Service Issue” was the most frequently reported complaint with a frequency of 2,910 tweets. This indicates that customer service problems were among the major factors contributing to negative traveller experiences. The visualization of airline sentiments distribution showed negative tweets appeared most frequently, followed by neutral and positive tweets respectively where positive tweets represented the smallest sentiment category. The visualization of negative reasons distribution showed that “Customer Service Issue” was the most common negative reason among traveller tweets, followed by “Late Flight” and “Can’t Tell”. Whereas “Damaged Luggage” and “longlines” appeared least frequently among the reported negative reasons. After preprocessing, the dataset shape became 14,604 records and 16 columns. Also, duplicate records were re-checked again, resulting in zero duplicate observations after cleaning. Regarding missing values, most missing observations did not directly affect the main sentiment analysis process, which mainly focused on tweet text and sentiment classification, therefore, missing values were not handled in this project. After that, sentiment analysis was performed and final sentiment analysis dataframe successfully stored cleaned tweets alongside their associated estimated sentiments. The displayed output demonstrated that the TextBlob library was successfully classify tweets into positive, negative, and neutral sentiment categories based on tweet polarity. This final dataframe can support easier interpretation and further analysis of traveller’s opinions toward airline services.

Conclusion

The results showed that negative tweets appeared more frequently than positive and neutral tweets, while customer service issues represented the most common negative reason among travelers. The sentiment analysis was successfully applied to estimate tweet sentiment

polarity and generate a final dataframe of sentiments results. Overall, this project demonstrates how sentiment analysis and text mining techniques can be used in real-world applications in order to better understand customer opinions and support service improvement.